

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12398889
Kind Code	B1
Date of Patent	August 26, 2025
Inventor(s)	Bailey, Jr.; James E

Furnace shut off device

Abstract

A furnace shut off device including a sensor assembly, a shutoff assembly and an alarm assembly. The sensor assembly includes a plurality of sensors. The plurality of sensors can be used to collect data of carbon monoxide concentration level. The shutoff assembly includes a microcontroller. The microcontroller receives the collected data from the plurality of sensors to shut off a furnace unit if the carbon monoxide concentration level is high. The alarm assembly includes a display, a light and a speaker. The microcontroller actuates said display, said light and said speaker if the carbon monoxide concentration level is high.

Inventors:	Bailey, Jr.; James E (Reedville, VA)
Applicant:	Bailey, Jr.; James E (Reedville, VA)
Family ID:	1000007532690
Appl. No.:	17/962693
Filed:	October 10, 2022

Publication Classification

Int. Cl.: F24D19/10 (20060101); F23N5/24 (20060101); G01N33/00 (20060101)

U.S. Cl.:

CPC F24D19/10 (20130101); F23N5/242 (20130101); G01N33/0063 (20130101);
F23N2231/22 (20200101)

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5896089	12/1998	Bowles	N/A	N/A
6989757	12/2005	Geoffrey J.	340/506	G08B 25/14
2013/0157559	12/2012	Flammer, III	455/7	G08B 21/12
2017/0261214	12/2016	Lai	N/A	H04L 12/00
2019/0221101	12/2018	Golob	N/A	H04W 4/80
2019/0226595	12/2018	Mattos	N/A	F17D 5/06
2019/0378399	12/2018	Hoofe, IV	N/A	G08B 25/10
2020/0372768	12/2019	Armpriester	N/A	G08B 21/02
2021/0145703	12/2020	Sjöo	N/A	A61Q 19/00
2021/0180235	12/2020	Gnadeberg	N/A	D06F 33/37
2021/0222877	12/2020	Combe	N/A	G08B 21/182
2021/0349066	12/2020	Chilla	N/A	G08B 21/14
2021/0407280	12/2020	Hoofe, IV	N/A	G08B 25/016
2022/0027725	12/2021	Nongpiur	N/A	G06N 7/01
2022/0043412	12/2021	Dedul	N/A	F24D 11/005

Primary Examiner: Lin; Ko-Wei

Attorney, Agent or Firm: Sanchelima & Associates, P.A.

Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to a furnace shut off device and, more particularly, to a furnace shut off device that can automatically shut off a furnace unit if a dangerous level of carbon monoxide (CO) is detected.

2. Description of the Related Art

(2) Several designs for shut off devices have been designed in the past. None of them, however, include a transmitter to send an alarm to a mobile device when dangerous CO levels are detected.

(3) Applicant believes that a related reference corresponds to U.S. Pat. No. 6,989,757 issued for a proactive carbon monoxide monitoring, alarm and protection system that is adapted to turn off the source of carbon monoxide, sound a central alarm, and alert an off-site monitoring station upon detection of carbon monoxide by the detector. Applicant believes that another related reference corresponds to U.S. Pat. No. 5,896,089 issued for a dual carbon monoxide detection system with gas cut off and alarm capabilities. None of these references, however, teach of a furnace safety device comprising multiple carbon monoxide sensors connected to an alarm system and an automatic shut off device, such that if the sensors measure an elevated or dangerous level of CO in the residence, the system will emit alarms and shut down the furnace.

(4) Other documents describing the closest subject matter provide for a number of more or less complicated features that fail to solve the problem in an efficient and economical way. None of these patents suggest the novel features of the present invention.

SUMMARY OF THE INVENTION

- (5) It is one of the objects of the present invention to provide a furnace shut off device that includes a speaker suitable to emit an audible sound to prevent the user of high CO levels.
- (6) It is another object of this invention to provide a furnace shut off device that includes a display to show the current CO level and prevent if that level is dangerous, the display may also display if the sensors are working adequately.
- (7) It is still another object of the present invention to provide a furnace shut off device that includes a microcontroller suitable to control the shut off of a furnace unit if the CO levels are high.
- (8) It is yet another object of this invention to provide such a device that is inexpensive to implement and maintain while retaining its effectiveness.
- (9) Further objects of the invention will be brought out in the following part of the specification, wherein detailed description is for the purpose of fully disclosing the invention without placing limitations thereon.
-

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) With the above and other related objects in view, the invention consists in the details of construction and combination of parts as will be more fully understood from the following description, when read in conjunction with the accompanying drawings in which:
- (2) FIG. 1 represents an isometric operational view of the present invention **10** wherein a user U receives an alarm indicating that the CO levels are in dangerous levels.
- (3) FIG. 2 shows an isometric view of the present invention **10**. The present invention **10** includes a sensor assembly **20**, a shutoff assembly **40** and an alarm assembly **60**.
- (4) FIG. 3 illustrates a diagram of the sensor assembly **20** connected to the shutoff assembly **40** and the shutoff assembly **40** connected to the alarm assembly **60** and the furnace unit **80**.
- (5) FIG. 4 is a representation of a diagram of the internal components of the shutoff assembly **40**.

DETAILED DESCRIPTION OF THE EMBODIMENTS OF THE INVENTION

- (6) Referring now to the drawings, where the present invention is generally referred to with numeral **10**, it can be observed that it basically includes a sensor assembly **20**, a shutoff assembly **40** and an alarm assembly **60**. It should be understood there are modifications and variations of the invention that are too numerous to be listed but that all fit within the scope of the invention. Also, singular words should be read as plural and vice versa and masculine as feminine and vice versa, where appropriate, and alternative embodiments do not necessarily imply that the two are mutually exclusive.
- (7) The sensor assembly **20** may include a plurality of sensors **22**, at least one port **24** and a battery **26**. As best illustrated in FIG. 2 and FIG. 4 the plurality of sensors may be located inside of the shutoff assembly **40**. It also may be suitable for the plurality of sensors **22** to be located outside of the shutoff assembly **40**. In an exemplary embodiment the plurality of sensors **22** may be located outside of the shutoff assembly **40** and be connected through at least one port **24**. The plurality of sensors **22** may be located on different locations in a building.
- (8) It should be understood that the plurality of sensors **22** may be carbon monoxide (CO) sensors. The plurality of sensors **22** may be a biomimetic sensor, an electrochemical sensor, a metal oxide semiconductor, or any other carbon monoxide sensor known in the prior art. The plurality of sensors **22** may sense data of a carbon monoxide concentration level. The port **24** may be located on a right portion of the shutoff assembly **40**. It also may be suitable for port **24** to have any other configuration in the shutoff assembly **40**. The plurality of sensors **22** may be connected to a battery **22**. The battery **22** may provide power to the plurality of sensors. It also may be suitable for the plurality of sensors **22** to be connected to a power outlet. The plurality of sensors **22** may be in communication with the shutoff assembly **40**.

(9) The shutoff assembly **40** may include a housing **42**, storage **43**, a microcontroller **41**, transmitter **45** an actuator **47**, a power cord **44** and a storage port **48**. The housing **42** may have a rectangular shape. It also may be suitable for the housing **42** to have a circular shape, a cylindrical shape, or any other suitable shape. The housing **42** may be made of plastic, metal, or any other suitable material. The housing **42** may enclose the storage **43**, the microcontroller **41**, the transmitter **45** the actuator **47**, the power cord **44** and the storage port. It also may be suitable for the housing **42** to enclose the sensor assembly **20** and the alarm assembly **60**. The housing **42** may be hollow.

(10) Referring now to FIG. **3** it can be observed that the plurality of sensors **22** may be connected to microcontroller **41**. Thus, data of carbon monoxide concentration level detected by the sensor **22** may be transmitted to the microcontroller **41**. The microcontroller **41** may identify if the carbon monoxide concentration level is high. If the carbon monoxide concentration level is high the microcontroller **41** may be actuated to shut off the furnace unit **80**. A high level of carbon monoxide is normally any concentration greater than 101 PPM. The microcontroller **41** may be powered through the power cord **44**. It also may be suitable for the microcontroller **41** to be powered through battery **26**. Battery **26** may be a lithium battery, a carbon-zinc battery, or any other suitable battery.

(11) The microcontroller **41** may control the transmitter **45**. In an exemplary embodiment the microcontroller **41** may control the transmitter **45** to send a signal to a mobile device **m** if the carbon monoxide concentration level is greater than 101 PPM. The mobile device **m** may display an alert or emit a sound to indicate that the carbon monoxide concentration level is high. The transmitter **45** may preferably use Bluetooth technologies. It also may be suitable for the transmitter **45** to be used to send collected data from the plurality of sensors **22** to the mobile device **m** so the mobile device can store the collected data.

(12) It also may be suitable to store the collected data from the plurality of sensors **22** through the storage **43**. In an exemplary embodiment the storage **43** is a removable storage that can be inserted into storage port **48**. Storage **43** may be a memory card, a pen drive or any other suitable storage known in the prior art.

(13) The microcontroller **41** may be connected to actuator **41**. The actuator **41** may be used to actuate the alarm assembly **60** to warn if the carbon monoxide concentration level is high. The shutoff assembly **40** may warn if the carbon monoxide concentration level is high through the transmitter **45** or through the actuator **47**. The shutoff assembly **40** may be used to shut off the furnace unit **80** if carbon monoxide concentration level is high.

(14) The alarm assembly **60** may include at least one speaker **62**, at least one light **64** and a display **61**. The at least one speaker **62** may be located on a front portion of the housing **42**. It also may be suitable for the at least one speaker **62** to have any other configuration in the housing **42**. In an alternative embodiment the speaker **62** may be placed separately from the housing **42**. The at least one speaker **62** may emit a predetermined sound. The predetermined sound may indicate that carbon monoxide concentration level is high. The at least one light **64** may be located on a front portion of the housing **42**. It also may be suitable for the at least one light **64** to have any other configuration in the housing **42**. The at least one light **64** may preferably be a light emitting diode. The at least one light **64** may turn on if the carbon monoxide concentration level is high. The at least one light **64** may blink if the carbon monoxide concentration level is high.

(15) The display **61** may also be located on a front wall of the housing **42**. The display **61** may display the carbon monoxide concentration level. The display **61** may also display if the furnace unit **80** is on or off. The display **61** may also display the status of the plurality of sensors **22**. The display **61** may further display an alert sign if the carbon monoxide concentration level is high. Display **61** may be located on a front wall of the housing **22**. It also may be suitable for the display **61** to have any other configuration in the housing **42**. The display **61** may be a liquid crystal display, a thin-film transistor, a light emitting diode display, or any other display known in the prior art. The display **61**, the at least one speaker **62** and the at least one light **64** may be actuated through

the actuator 47 to warn if the carbon monoxide concentration level is high.

(16) The foregoing description conveys the best understanding of the objectives and advantages of the present invention. Different embodiments may be made of the inventive concept of this invention. It is to be understood that all matter disclosed herein is to be interpreted merely as illustrative, and not in a limiting sense.

Claims

1. A furnace shut off device, comprising: a sensor assembly, the sensor assembly includes a plurality of sensors, wherein said plurality of sensors are carbon monoxide sensors, said sensors are configured to collect data of carbon monoxide concentration level; a shutoff assembly, wherein said shutoff assembly includes a housing and a microcontroller, said housing has a rectangular shape, said microcontroller is inside said housing, said microcontroller is in communication with said plurality of sensors, said microcontroller is programmed to shut off a furnace unit if said carbon monoxide concentration level is greater than 101 parts per million; and an alarm assembly, wherein said alarm assembly is in communication with said shutoff assembly, said alarm assembly includes at least one light, at least one speaker and a display, said at least one light, said at least one speaker, and said display being located on a front wall of said housing, wherein said at least one light is turned on if said carbon monoxide concentration level is greater than 101 parts per million, said speaker emits a predetermined sound if said carbon monoxide concentration level is greater than 101 parts per million, said display is configured to display a warning signal if said carbon monoxide concentration level is greater than 101 parts per million.
2. The furnace shut off device set forth in claim 1, wherein said plurality of sensors are inside said housing.
3. The furnace shut off device set forth in claim 1, wherein said plurality of sensors are connected to said shutoff assembly by means of a cord, wherein said cord is capable of being connected to a port in said housing.
4. The furnace shut off device set forth in claim 1, wherein said plurality of sensors are connected to a battery, said battery provides power to said plurality of sensors.
5. The furnace shut off device set forth in claim 1, wherein said shutoff assembly further includes a transmitter, said transmitter is adapted to transmit a warning signal to a mobile device.
6. The furnace shut off device set forth in claim 5, wherein said transmitter uses Bluetooth technologies.
7. The furnace shut off device set forth in claim 1, wherein said shutoff assembly includes an actuator, wherein said actuator actuates said speaker to sound, said actuator actuates said light to blink, said actuator is controlled by said microcontroller.
8. The furnace shut off device set forth in claim 1, wherein said display is adapted to display a first status for said plurality of sensors and a second status for said furnace unit, said first status is on or off, said second status is on or off.
9. The furnace shut off device set forth in claim 1, wherein said shutoff assembly includes storage, wherein said storage is capable of being used to store said collected data.
10. A furnace shut off device, comprising: a sensor assembly, the sensor assembly includes a plurality of sensors, wherein said plurality of sensors are carbon monoxide sensors, said sensors are configured to collect data of carbon monoxide concentration level; a shutoff assembly, wherein said shutoff assembly includes a housing, a transmitter, and a microcontroller, said housing having a rectangular shape, said microcontroller is inside said housing, said microcontroller is in communication with said plurality of sensors, said microcontroller is adapted to shut off a furnace unit if said carbon monoxide concentration level is greater than 101 parts per million, said transmitter is adapted to transmit a warning sign to a mobile device; and an alarm assembly, wherein said alarm assembly is in communication with said shutoff assembly, said alarm assembly

includes at least one light, at least one speaker and a display, said at least one light, said at least one speaker, and said display being located on a front wall of said housing, wherein said at least one light is turned on if said carbon monoxide concentration level is greater than 101 parts per million, said speaker emits a predetermined sound if said carbon monoxide concentration level is greater than 101 parts per million, said display is configured to display a warning signal if said carbon monoxide concentration level is greater than 101 parts per million, wherein said display is adapted to display a first status for said plurality of sensors and a second status for said furnace unit, said first status is on or off, said second status is on or off.

11. A furnace shut off device, consisting of: a sensor assembly, the sensor assembly includes a plurality of sensors, wherein said plurality of sensors are carbon monoxide sensors, said sensors are configured to collect data of carbon monoxide concentration level; a shutoff assembly, wherein said shutoff assembly includes a housing, storage, a transmitter, and a microcontroller, said housing having a rectangular shape, said microcontroller is inside said housing, said microcontroller is in communication with said plurality of sensors, said microcontroller is adapted to shut off a furnace unit if said carbon monoxide concentration level is greater than 101 parts per million, said transmitter is adapted to transmit a warning sign to a mobile device, said transmitter uses Bluetooth technology, wherein said storage is capable of be used to store said collected data, wherein said plurality of sensors are inside of said housing, said plurality of sensors are connected to a battery, said microcontroller provides power to said plurality of sensors, said battery provides emergency power to said microcontroller; and an alarm assembly, wherein said alarm assembly is in communication with said shutoff assembly, said alarm assembly includes at least one light, at least one speaker and a display, said at least one light, said at least one speaker, and said display being located on a front wall of said housing, wherein said at least one light is turned on if said carbon monoxide concentration level is greater than 101 parts per million, said speaker emits a predetermined sound if said carbon monoxide concentration level is greater than 101 parts per million, said display is configured to display a warning signal if said carbon monoxide concentration level is greater than 101 parts per million, wherein said display is adapted to display a first status for said plurality of sensors and a second status for said furnace unit, said first status is on or off, said second status is on or off.
